readme

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1 Introduction

1.1 Overview

This project implements a multi-client quiz application using TCP sockets in C++. It consists of two parts:

- **Part 1A:** Basic client-server system with LLM integration (no cache).
- **Part 1B:** Extended version with server-side caching of quiz questions per genre.

Clients can:

- Connect to the server
- Select a quiz genre
- Receive dynamically generated multiple-choice questions
- Submit answers and receive feedback
- View a real-time leaderboard
- Handle keep-alive probes from the server

1.2 Setup

- Operating System: Ubuntu 22.04 LTS (tested)
- Compiler: g++ (version $\geq$ 9.4.0) with C++17 support
- Dependencies:
  - `nlohmann/json.hpp` (header-only, included)
  - Python 3.x (for tester script)

1.3 Compilation

Part 1A

```
g++ -std=c++17 -pthread server.cpp -o server
g++ -std=c++17 -pthread client.cpp -o client
```

Part 1B

```
g++ -std=c++17 -pthread server_B.cpp -o server_B
g++ -std=c++17 -pthread client_B.cpp -o client_B
```

1.4 Running

Start the Server (Default port: 11043)

```
./server      # Part 1A
./server_B    # Part 1B
```

Start Clients

```
./client <server_ip> 11043      # Part 1A client
./client_B <server_ip> 11043    # Part 1B client
```

Example:

```
./client 192.168.50.111 11043
```

1.5 Expected Input/Output

Sample Client Run

```
Connected to the quiz server!
Please enter a quiz genre (e.g., science, history, movies): science

------------------------
Question: Who proposed the theory of relativity?
A) Isaac Newton
B) Albert Einstein
C) Galileo Galilei
D) Nikola Tesla
------------------------
Your answer (A, B, C, D) or type 'leaderboard': B

>>> Result: CORRECT | Your Score: 10
Waiting for the next question...
```

Leaderboard Example

```
--- Leaderboard ---
Client 0: 20
Client 1: 10
Client 2: 0
-------------------
Your answer (A, B, C, D) or type 'leaderboard':
```

Cache Mode (Part 1B)

```
(Question from cache)
Question: Who discovered penicillin?
A) Louis Pasteur
B) Alexander Fleming
C) Marie Curie
D) Joseph Lister
```

1.6 Automated Testing (tester.py)

A Python tester simulates multiple clients.
Usage:

```
python3 tester.py <server_ip> 11043
```

The tester prompts for the number of clients to launch. Each client:

- Randomly selects a genre
- Randomly answers questions
- Handles server responses automatically

1.7 Performance analysis

PART A

Concurrent Clients	Avg. Initial Question Latency (ms)	Avg. Answer Latency (ms)	Avg. Throughput per Client (KB/s)
1	855.4	1.2	5.12
3	2410.8	3.5	1.85
5	4150.2	5.8	1.11
7	5585.1	8.1	0.83
10	8233.6	11.4	0.56

PART B

Concurrent Clients	Avg. Initial Question Latency (ms)	Avg. Answer Latency (ms)	Avg. Throughput per Client (KB/s)
1	861.2	1.3	5.09
3	295.7	1.4	14.81
5	179.9	1.5	15.15
7	132.1	1.6	15.24
10	95.3	1.8	15.31

Figure 1: performance analysis values by using pcap files